The documentation and process conversion measures necessary to comply with this document shall be completed by 22 August 2009.

INCH-POUND

MIL-PRF-19500/464G
22 May 2009
SUPERSEDING
MIL-PRF-19500/464F
27 August 2007

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER,
TYPES 2N5685 AND 2N5686, JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN silicon, power transistors. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1, similar to TO-3.

1.3 Maximum ratings. T_C = +25°C, unless otherwise specified.

Type	P_T (1) T_C = +25°C	P_T (1) T_C = +100°C	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_J and T_{STG}	$R_{\theta JC}$
	W	W	V dc	V dc	V dc	A dc	A dc	°C	°C/W
2N5685	300	171	60	60	5	15	50	-55 to +200	.584
2N5686	300	171	80	80	5	15	50	-55 to +200	.584

(1) Between T_C = +25°C and T_C = +200°C linear derating factor 1.715 W/°C.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.

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1.4 Primary electrical characteristics. Unless otherwise specified, T_C = +25°C.

Type	h_{FE2} (1)		h_{FE3} (1)		$V_{BE(sat)}$ (1)		$V_{CE(sat)1}$ (1)		$V_{CE(sat)2}$ (1)	
	V_{CE} = 2 V dc I_C = 25 A dc		V_{CE} = 5 V dc I_C = 50 A dc		I_C = 25 A dc I_B = 2.5 A dc		I_C = 25 A dc I_B = 2.5 A dc		I_C = 50 A dc I_B = 10 A dc	
	Min	Max	Min	Max	V dc Min	Max	V dc Min	Max	V dc Min	Max
2N5685	15	60	5			2.0		1.0		5.0
2N5686	15	60	5			2.0		1.0		5.0

Type	C_{obo}		$\|h_{fe}\|$		h_{fe}		Switching (see table I and figure 2 herein)	
	V_{CB} = 10 V dc I_E = 0 0.1 MHz ≤ f ≤ 1.0 MHz		V_{CE} = 10 V dc I_C = 5 A dc f = 1 MHz		V_{CE} = 5 V dc I_C = 10 A dc f = 1 kHz		t_{on}	t_{off}
	Min	Max pF	Min	Max	Min	Max	Min Max µs	Min Max µs
2N5685		1,200	2	20	15		1.5	3.0
2N5686		1,200	2	20	15		1.5	3.0

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

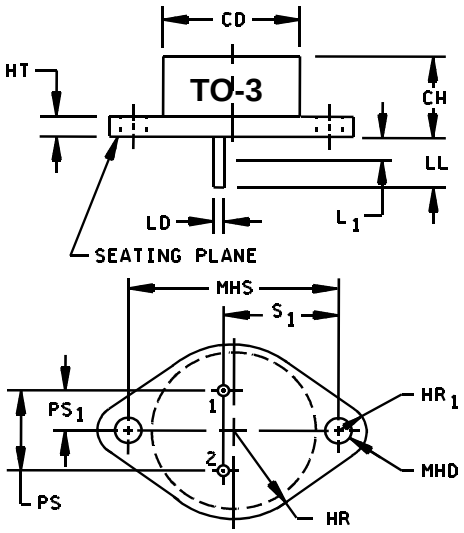
DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD		.875		22.22	3
CH	.250	.450	6.35	11.43	
HR	.495	.525	12.57	13.34	
HR_1	.131	.188	3.33	4.78	6
HT	.060	.135	1.52	3.43	
LD	.057	.063	1.45	1.60	4, 5, 9
LL	.312	.500	7.92	12.70	4, 5, 9
L_1		.050		1.27	5, 9
MHD	.151	.165	3.84	4.19	7
MHS	1.177	1.197	29.90	30.40	
PS	.420	.440	10.67	11.18	
PS_1	.205	.225	5.21	5.72	5
S_1	.655	.675	16.64	17.15	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Body contour is optional within zone defined by CD.
4. These dimensions shall be measured at points .050 inch (1.27 mm) to .055 inch (1.40 mm) below seating plane. When gauge is not used, measurement shall be made at seating plane.
5. Both terminals.
6. At both ends.
7. Two holes.
8. The collector shall be electrically connected to the case.
9. LD applies between L_1 and LL. Lead diameter shall not exceed twice LD within L_1.
10. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.
* 11. Terminal 1 is emitter; terminal 2 is base; case is collector.

* FIGURE 1. Physical dimensions.

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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* 4.3 Screening (JANTX and JANTXV levels). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500)	Measurement JANTX and JANTXV levels only
(1) 3c	Thermal impedance (see 4.3.2)
11	h_{FE2} and I_{CEX1}
12	See 4.3.1
13	Subgroup 2 of table I herein; ΔI_{CEX1} = 100 percent of initial value or 1 μA dc, whichever is greater; Δh_{FE2} = 25 percent of initial value.

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_J = +187.5°C ±12.5°C; $V_{CB} \geq$ 20 V dc, $T_A \leq$ +100°C.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW}, t_{MD} and V_H where appropriate). See table II, group E, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, appendix E table E-V, and table I herein. Electrical measurements (end-points) shall be in accordance with the table I, subgroup 2 herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the table I, subgroup 2 herein.

Subgroup	Method	Conditions
B3	1037	2,000 cycles, $V_{CE} \geq$ 10 V dc, ΔT_J between cycles ≥ +100°C, adjust power or current to achieve a ΔT_J = +100°C.

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* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the table I, subgroup 2 herein.

Subgroup	Method	Conditions
C2	2036	Test condition A, weight = 10 pounds, t = 15 s.
C5	3131	See 4.3.2, $R_{\theta JC}$ = .584°C/W.
C6	1037	6,000 cycles, $V_{CE} \geq$ 10 V dc, ΔT_J between cycles $\geq$ +100°C, adjust power or current to achieve a ΔT_J = +100°C.

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified herein. Electrical measurements (end-points) shall be in accordance with the table I, subgroup 2 herein.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance 2/	3131	See 4.3.2	$Z_{\theta JX}$			°C/W
Breakdown voltage collector to emitter	3011	Bias condition D; I_C = 100 mA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N5685				60		V dc
2N5686				80		V dc
Collector to emitter cutoff current	3041	Bias condition D	I_{CEO}		500	µA dc
2N5685		V_{CE} = 30 V dc				
2N5686		V_{CE} = 40 V dc				
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = 1.5 V dc	I_{CEX1}		10	µA dc
2N5685		V_{CE} = 60 V dc				
2N5686		V_{CE} = 80 V dc				
Emitter to base cutoff current	3061	Bias condition D; V_{BE} = 5 V dc, I_C = 0	I_{EBO}		1.0	mA dc
Collector to base cutoff current	3036	Bias condition D	I_{CBO1}		2.0	mA dc
2N5685		V_{CE} = 60 V dc				
2N5686		V_{CE} = 80 V dc				
Base to emitter saturated	3066	Test condition A; I_C = 25 A dc, I_B = 2.5 A dc, pulsed (see 4.5.1)	$V_{BE(sat)}$		2.0	V dc
Base to emitter non-saturated	3066	Test condition B; I_C = 25 A dc, V_{CE} = 2 V dc, pulsed (see 4.5.1)	V_{BE}		2.0	V dc
Collector to emitter saturated voltage	3071	I_C = 25 A dc; I_B = 2.5 A dc pulsed (see 4.5.1)	$V_{CE(sat)1}$		1.0	V dc
Collector to emitter saturated voltage	3071	I_C = 50 A dc; I_B = 10 A dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		5.0	V dc
Forward current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 5 A dc, pulsed (see 4.5.1)	h_{FE1}	30		

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 25 A dc, pulsed (see 4.5.1)	h_{FE2}	15	60	
Forward current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 50 A dc, pulsed (see 4.5.1)	h_{FE3}	5		
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current 2N5685 2N5686	3041	Bias condition A; V_{BE} = 1.5 V dc V_{CE} = 60 V dc V_{CE} = 80 V dc	I_{CEX2}		5	mA dc
Low temperature operation:		T_A = -55°C				
Forward current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 25 A dc, pulsed (see 4.5.1)	h_{FE4}	7		
Subgroup 4						
Pulse response	3251	Test condition A, except test circuit and pulse requirements (see figure 2)				
Turn-on time		V_{CC} = 30 V dc; I_C = 25 A dc, I_{B1} = 2.5 A dc	t_{on}		1.5	µs
Turn-off time		V_{CC} = 30 V dc; I_C = 25 A dc, I_{B1} = - I_{B2} = 2.5 A dc	t_{off}		3.0	µs
Storage time		V_{CC} = 30 V dc; I_C = 25 A dc, I_{B1} = - I_{B2} = 2.5 A dc	t_s		2.0	µs
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 5 A dc, f = 1 MHz	$\|h_{fe}\|$	2	20	

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 5 V dc; I_C = 10 A dc, f = 1 kHz	h_{fe}	15		
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0, 0.1 MHz ≤ f ≤ 1 MHz	C_{obo}		1,200	pF
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +25°C; t = 1 s, 1 cycle (see figures 3 and 4)				
Test 1		V_{CE} = 6 V dc; I_C = 50 A dc				
Test 2		V_{CE} = 30 V dc; I_C = 10 A dc				
Test 3 2N5685		V_{CE} = 50 V dc; I_C = 560 mA dc				
2N5686		V_{CE} = 60 V dc; I_C = 640 mA dc				
Safe operating area (switching)	3053	Load condition C (unclamped inductive load) (see figure 5) T_C = +25°C duty cycle ≤ 10 percent R_S = 0.1 Ω; $t_r = t_f \leq$ 500 ns				
Test 1		t_p approximately 5 ms (vary to obtain I_C); R_{BB1} = 10 Ω; V_{BB1} = 20 V dc; R_{BB2} = ∞; V_{BB2} = 0 V; V_{CC} = 50 V dc; I_C = 20 A dc; L = 1 mH; Sanford Miller CK - 50, 50 A .002 Ω (or equivalent)				
Test 2		t_p approximately 5 ms (vary to obtain I_C); R_{BB1} = 100 Ω; V_{BB1} = 10 V dc; R_{BB2} = ∞; V_{BB2} = 0 V; V_C = 50 V dc; I_C = 1.5 A dc; L = 80 mH; (2 each signal transformer CH06, 6A) 0.4 Ω (or equivalent)				

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued						
Safe operating area (switching)	3053	Clamped inductive load (see figures 6 and 7) T_A = +25°C; V_{CC} = 50 V dc				
2N5685		Clamp voltage = 60 V dc				
2N5686		Clamp voltage = 80 V dc				
Electrical measurements		See table I, subgroup 2				

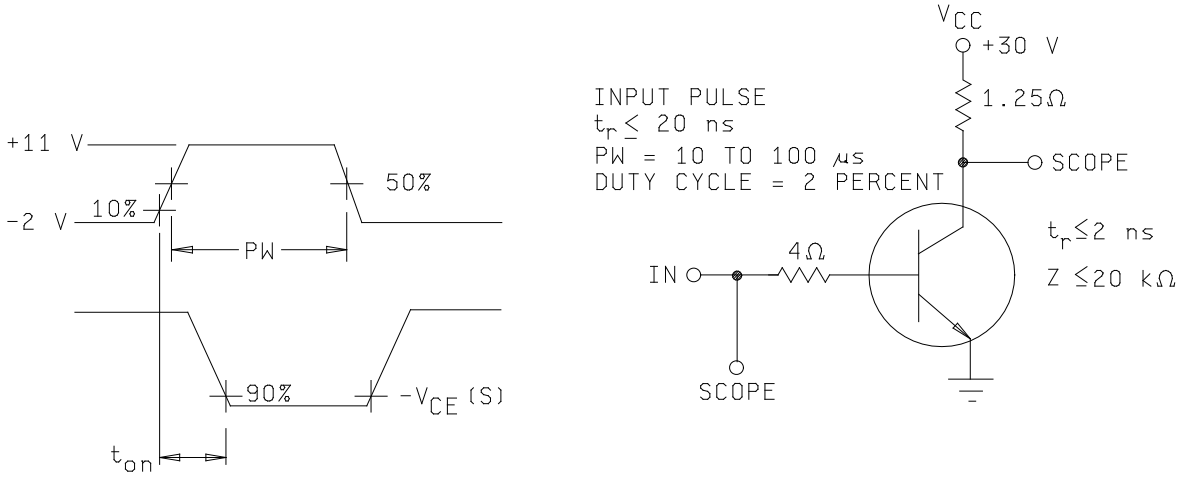
1/ For sampling plan, see MIL-PRF-19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroups 2 and 6.
Group E, subgroup 1.

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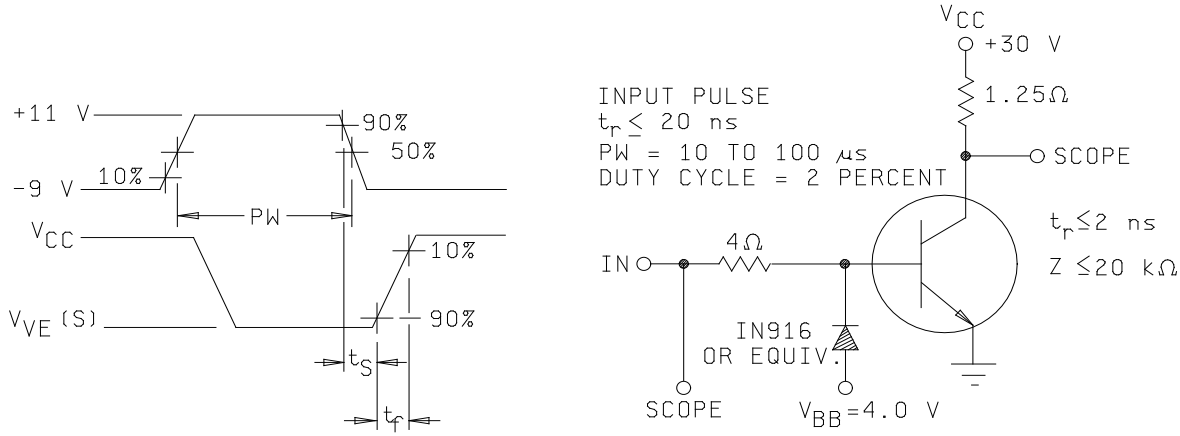
* TABLE II. Group E inspection (all quality levels) - for qualification or re-qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling	1051	Test condition C, 500 cycles.	
Hermetic seal Fine leak Gross leak	1071	 Test conditions G or H Test conditions C or D	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2			45 devices c = 0
Blocking life	1048	Test temperature = +125°C; V_{CB} = 30 V dc; T = 1,000 hours.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 4			
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 6			3 devices
ESD	1020		
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B.	

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TURN-ON (t_{ON}) TIME TEST CIRCUIT



toff = ts + tf

TURN-OFF (t_{OFF}) TIME TEST CIRCUIT

FIGURE 2. Switching time test circuits.

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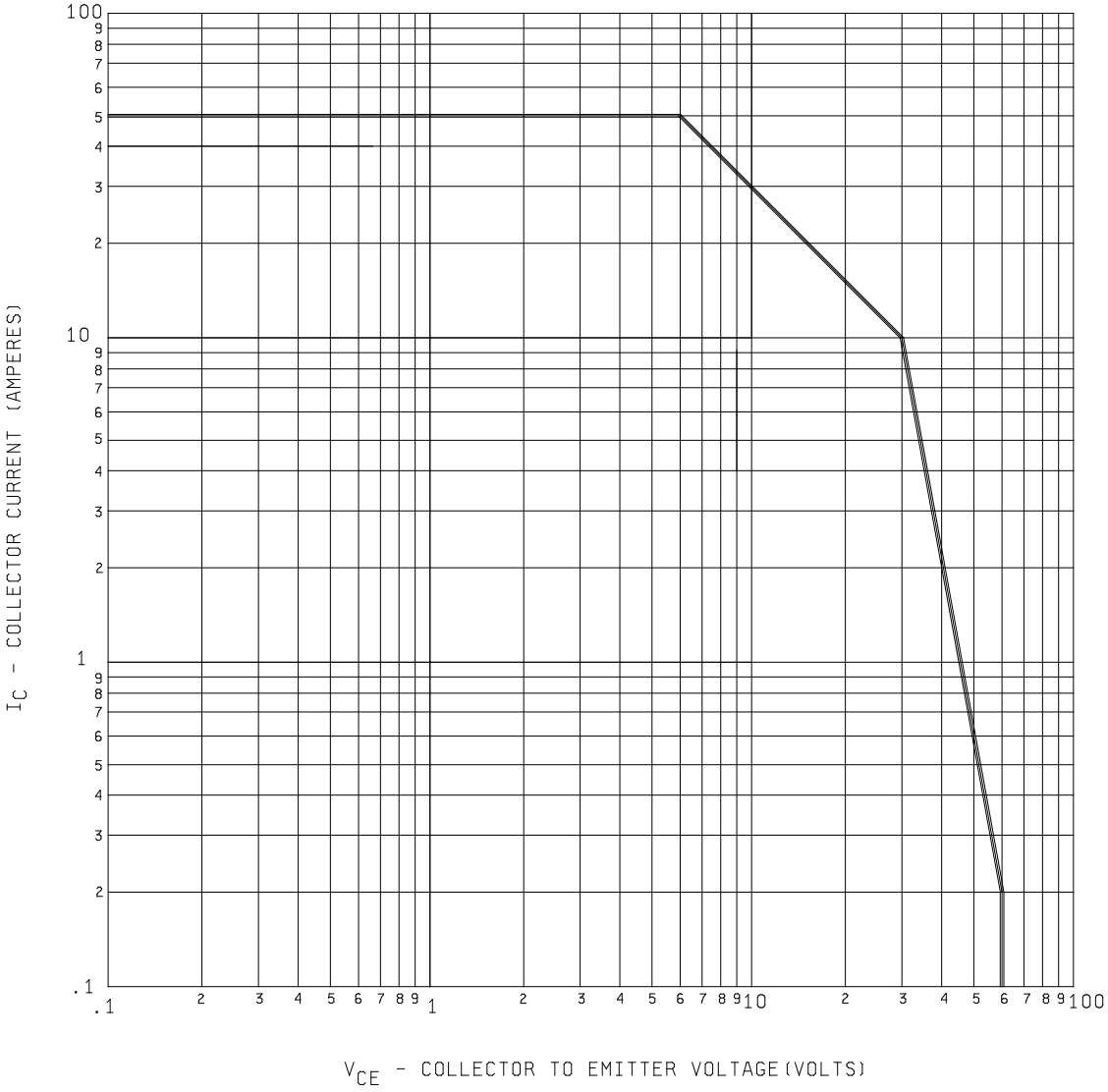


FIGURE 3. Maximum safe operating area graph continuous dc (2N5685).

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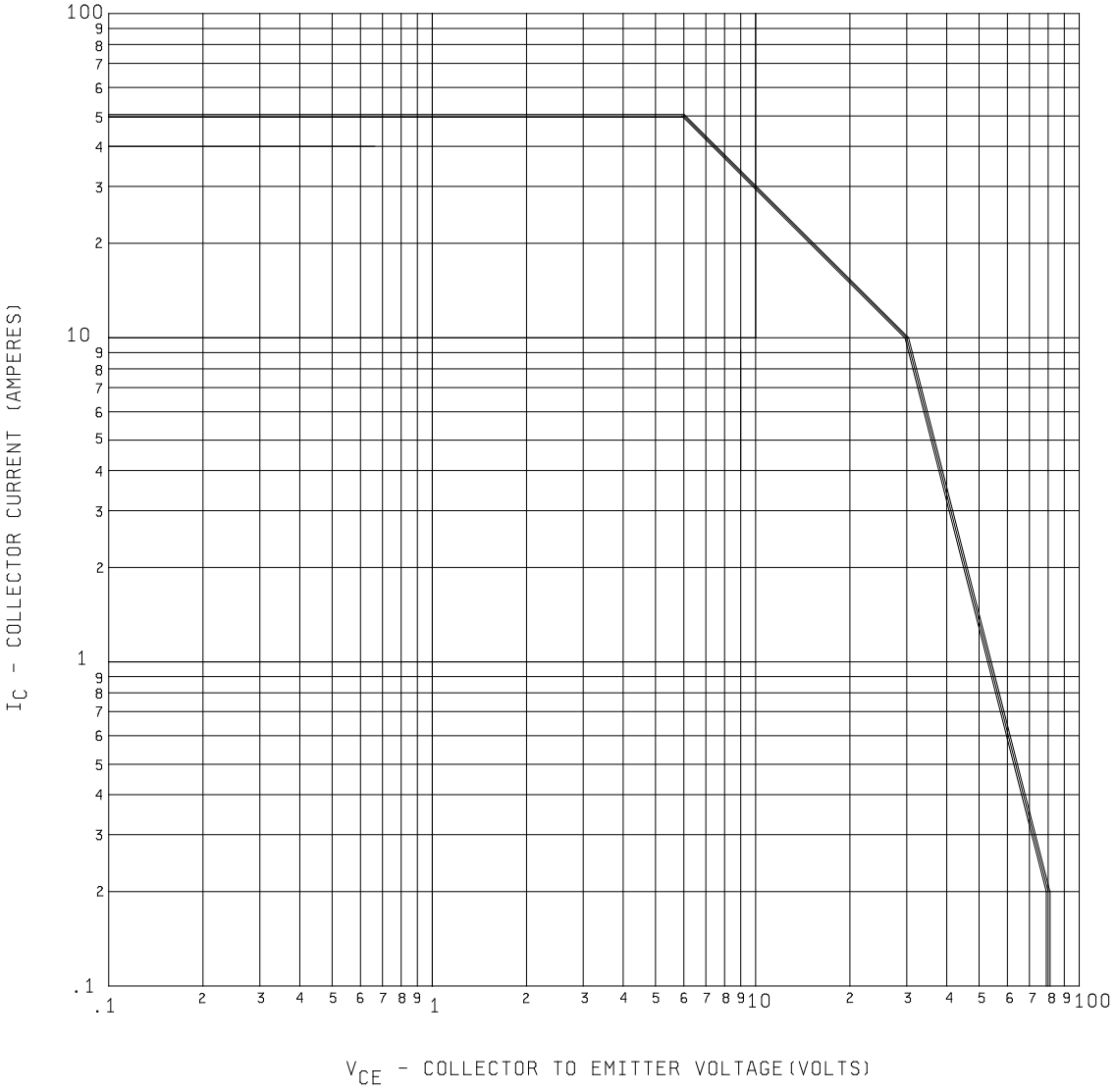


FIGURE 4. Maximum safe operating area graph continuous dc (2N5686).

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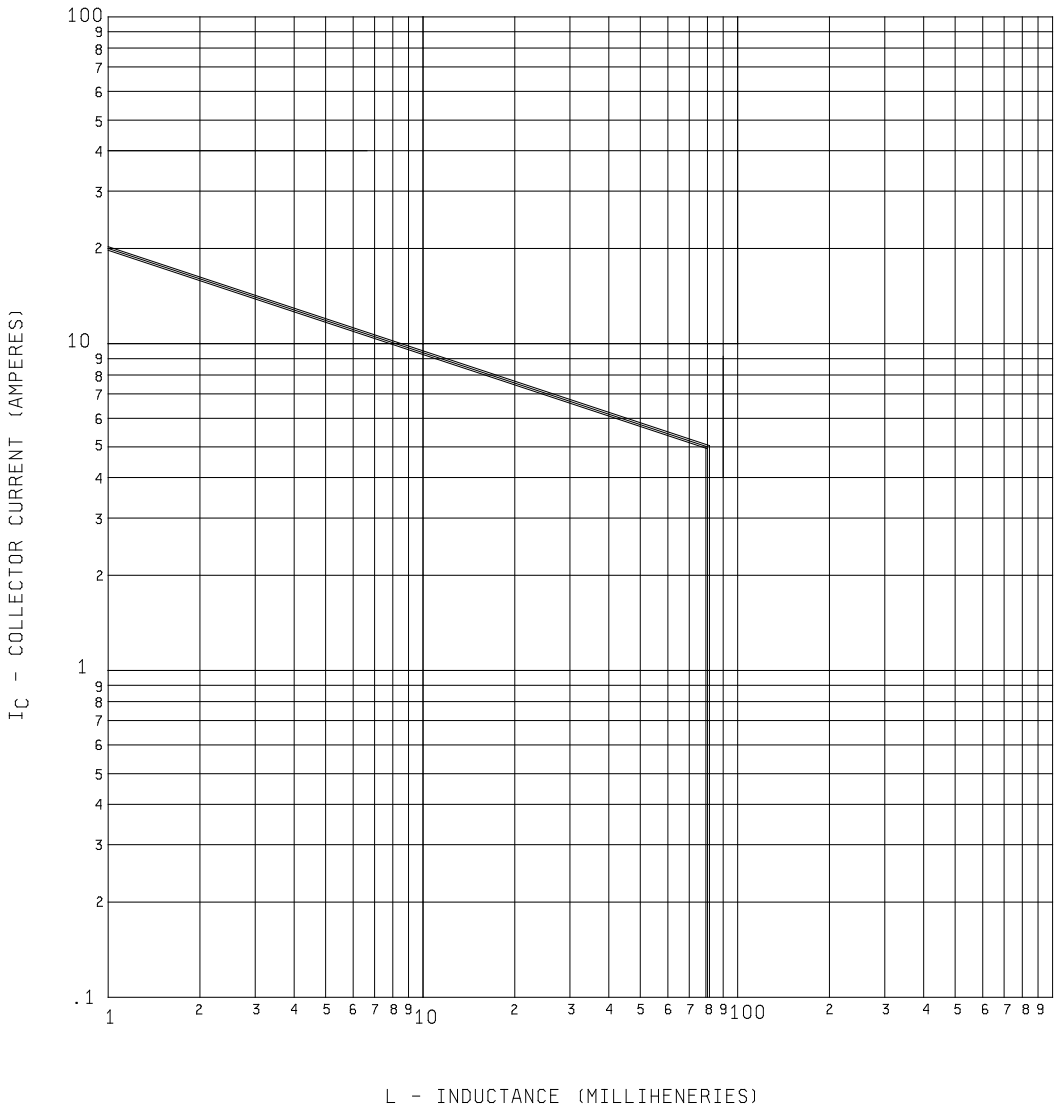
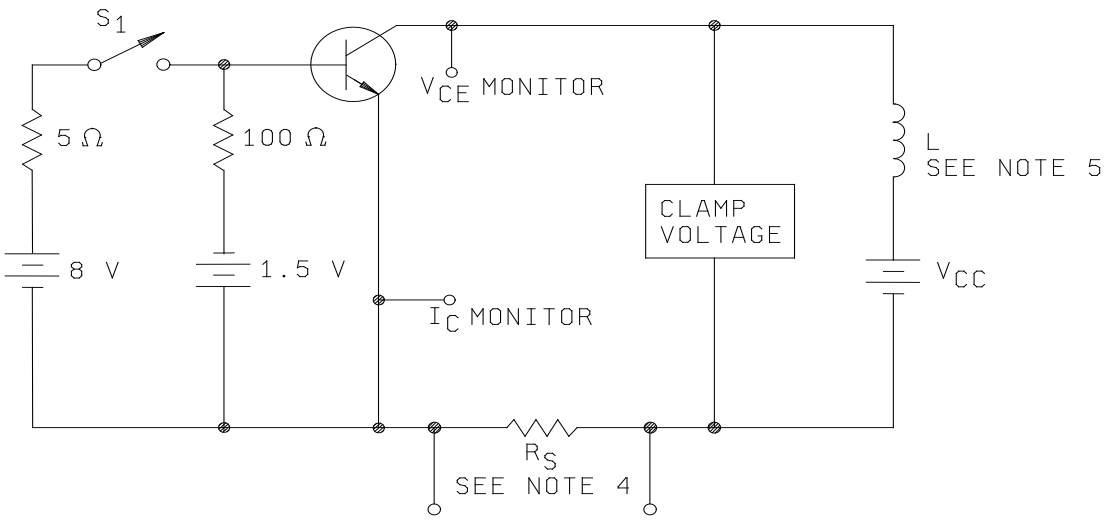


FIGURE 5. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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Procedure:

1. With switch S_1 closed, set the specified test conditions.
2. Open S_1. Device fails if clamp voltage not reached.
3. Perform specified end-point tests.
4. $R_S \leq 0.1\Omega$, 12 W; 1percent tolerance maximum; (noninductive)
5. L = 2.0 mH (2 each 1 mH. Sanford Miller CK-50, 50 A). R = .002 Ω.

FIGURE 6. Clamp inductive sweep test circuit.

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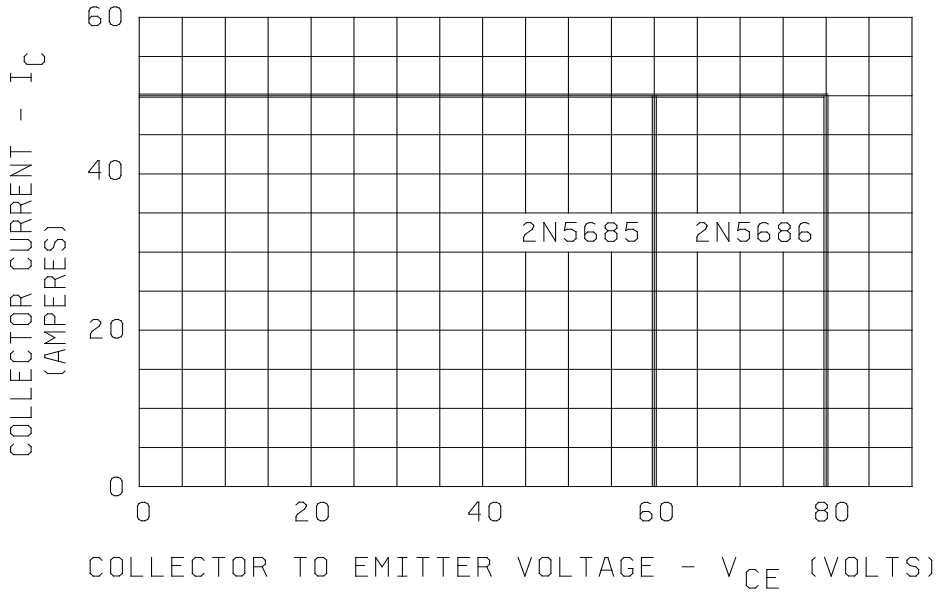


FIGURE 7. Safe operating area for switching between saturation and cutoff (clamped inductive load).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at http://assist.daps.dla.mil .

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* 6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AR, MI
Air Force - 19, 70, 99

Preparing activity:
DLA - CC

(Project 5961-2008-121)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/ .

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